Program : **Diploma in Polymer Technology**	
Course Code : **6092B**	Course Title: **Product Design and Development**
Semester : **6**	Credits: **4**
Course Category: **Program Elective**	
Periods per week: **4 (L:3, T:1, P:0)**	Periods per semester: **75**

Course Objectives:

- To acquire the basic concepts of product design and development process
- To understand the engineering and scientific process in executing a design from concept to finished product
- To study the key reasons for design or redesign.

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Course Prerequisites: Nil

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Identify the basic concepts of product design and development process.	15	Understanding
CO2	Illustrate the methods to define the customer needs.	14	Understanding
CO3	Describe an engineering design and development process.	14	Understanding
CO4	Apply modeling and embodiment principles in product design and development process by interpreting the intuitive and advanced methods used to develop and evaluate a concept	15	Applying
	Series test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2			3				
CO3					3		
CO4						3	

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Understand the basic concepts of product design and development process.**		
M1.01	To understand the Definition of a product, levels of product, product - market mix;	3	Understanding
M1.02	To comprehend the process of New product development (NPD)	4	Understanding
M1.03	To understand the Idea generation methods, Creativity, Creative attitude and Creative design process	4	Understanding
M1.04	Analysis of interconnected decision areas and brain storming.	4	Understanding
Contents: Definition of a product; Types of product; Levels of product; Product-market mix; New product development (NPD) process; Idea generation methods; Creativity; Creative attitude; Creative design process; Morphological analysis; Analysis of interconnected decision areas; Brain storming			
CO2	**Illustrate the methods to define the customer needs.**		
M2.01	To understand product life cycle, the challenges of product development and product analysis	3	Understanding
M2.02	To identify the product characteristics; economic considerations	3	Understanding
M2.03	To understand the production and marketing aspects; phases of a generic product development process	3	Understanding

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<tr><td>M2.04</td><td>To comprehend the characteristics of successful product development;</td><td>3</td><td>Understanding</td></tr>
<tr><td>M2.05</td><td>To understand customer need identification, product development practices and industry-product strategies.</td><td>2</td><td>Understanding</td></tr>
<tr><td></td><td>Series test I</td><td>1</td><td></td></tr>
<tr><td colspan="4">Contents:
Product life cycle; The challenges of Product development; Product analysis; Product characteristics; Economic considerations; Production and Marketing aspects; Characteristics of successful Product development; Phases of a generic product development process; Customer need identification; Product development practices and industry - product strategies.</td></tr>
<tr><td>CO3</td><td colspan="3">Describe an engineering design and development process.</td></tr>
<tr><td>M3.01</td><td>To understand the methods of product design-design by evolution, design by innovation and design by imitation</td><td>3</td><td>Understanding</td></tr>
<tr><td>M3.02</td><td>To identify the factors affecting product design- standards of performance and environmental factors;</td><td>3</td><td>Understanding</td></tr>
<tr><td>M3.03</td><td>To know Decision making and iteration;</td><td>4</td><td>Understanding</td></tr>
<tr><td>M3.04</td><td>To understand the Morphology of design (different phases) and Role of aesthetics in design.</td><td>4</td><td>Understanding</td></tr>
<tr><td colspan="4">Contents:
Product design; Design by evolution; Design by innovation; Design by imitation; Factors affecting product design; Standards of performance and environmental factors; Decision making and iteration; Morphology of design (different phases); Role of aesthetics in design.</td></tr>
<tr><td>CO4</td><td colspan="3">Understand the intuitive and advanced methods used to develop and evaluate a concept. Apply modeling and embodiment principles in product design and development process</td></tr>
<tr><td>M4.01</td><td>To understand optimization in design, economic factors in design and design for safety and reliability</td><td>2</td><td>Understanding</td></tr>
<tr><td>M4.02</td><td>To understand the Role of computers in design; Modeling and Simulation The role of models in engineering design; Mathematical modeling; Similitude and scale models</td><td>2</td><td>Understanding</td></tr>
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M4.03	To understand the concept of Six sigma and design for six sigma and Introduction to optimization in design for Economic factors and financial feasibility in design	2	Understanding
M4.04	To understand the design for manufacturing by rapid prototyping (RP) and its application of RP in product design	3	Applying
M5.04	To understand the design of simple products dealing with various aspects of product development; design starting from need till the manufacture of the product	3	Applying
M6.05	To comprehend about product development versus product design	3	Applying
	Series test II	1	

Contents:

Introduction to optimization in design; Economic factors in design; Design for safety and reliability; Role of computers in design; Modeling and Simulation; The role of models in engineering design; Mathematical modeling; Similitude and scale models; Concurrent design; Six sigma and design for six sigma; Introduction to optimization in design; Economic factors and financial feasibility in design; Design for manufacturing; Rapid Prototyping (RP); Application of RP in product design; Product Development versus Design. Design of simple products dealing with various aspects of product development; Design starting from need till the manufacture of the product

Text / Reference:

T/R	Book Title/Author
T1	Product Design and Development, Karl T. Ulrich and Steven D. Eppinger, Tata McGraw–Hill edition.
R2	Engineering Design –George E. Dieter.
R3	An Introduction to Engineering Design methods Vijay Gupta.
R4	Merie Crawford : New Product management, McGraw-Hill Irwin.

Online resources:

Sl. No	Website Link
1	https://www.smashingmagazine.com/ www.renfrewgroup.com - Product Design Services www.renfrewgroup.com - Product Design Services
2	https://www.interaction - design.org

3	https://en.wikipedia.org/wiki/Product_design
4	https://www.udacity.com